UTA Trax TPSS

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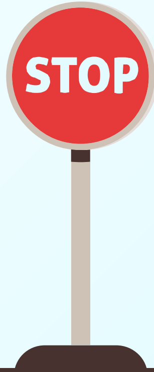


01 Background

Introduction

The Problem: The UTA (Utah Transit Authority) wants to build a new light rail station for the purpose of testing repairs on current LRVs (Light Rail Vehicle) or new LRVs to be used in the rest of the light transit network.

Our team will design a new TPSS (Traction Power Substation) to bid on the project.

This will be a 30% preliminary design that UTA engineers will review and confirm before our team will complete the 100% design.

TPSS

What is a TPSS?

A traction power substation is a purpose built distribution station that is meant to power electric trains in public transit. For our purpose, it will consist of these components:

- Input Breakers
- Rectifier Transformer
- Rail Feeders
- Protection Devices

Why is a TPSS needed?

LRVs use DC

The light rail vehicles will use 750 Volt DC as the input voltage.

The Grid is AC

The grid operates at variable AC voltages. The lines that service the station will be at 34.5 KV.

Initial Design Requirements

Light Rail Vehicle

We will be using 3 cars of the image shown below

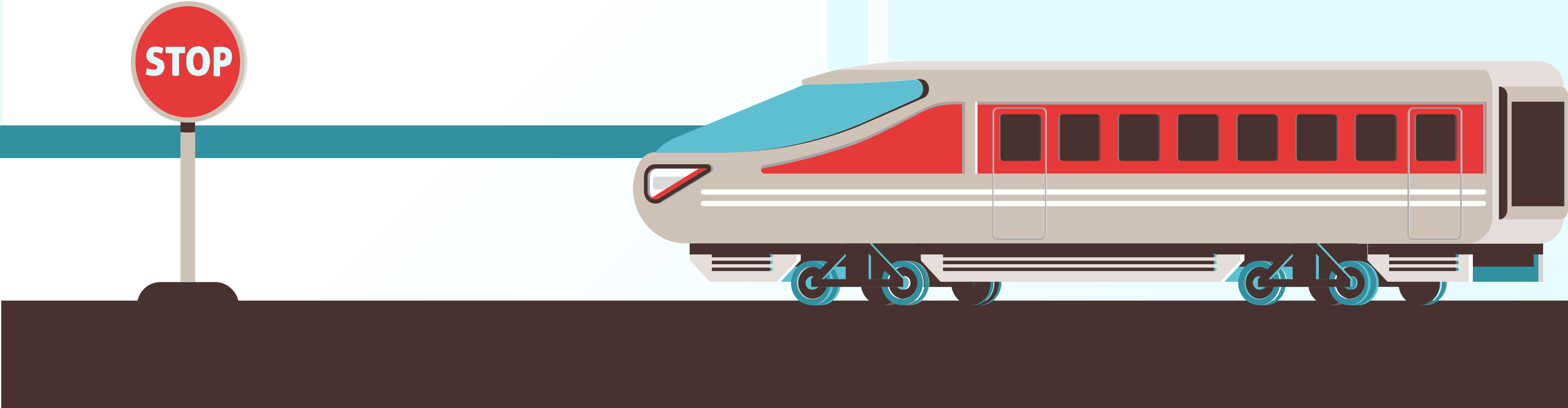
SB and NB

Two rails, one going South Bound and the other going North Bound

Accelerating/Coasting

Acceleration will be for 5 seconds and then the LRV will coast for 25 seconds at 10 mph

02 Results



Major Components Needed

Rectifier Transformer

Transforms and Rectifies 34.5 kV AC to 750 V DC

Protection Devices

Breakers, Fuses, Relays, etc.

Utility Feed

34.5 kV power lines that supplies TPSS stations

Overhead Catenary

Overhead cable that supplies power to the LRV

Protection Devices

AC/DC Breakers

Isolate affected portions

AC Breaker: Protects Metering Devices & Rectifier Transformer

DC Breaker: Protects OCS

MFPR

Multi-Function Protective Relays detect faults, overcurrents, over/under voltages, trips breakers, and contains other relays

Fuses

Provides simple backup protection
No control signal required

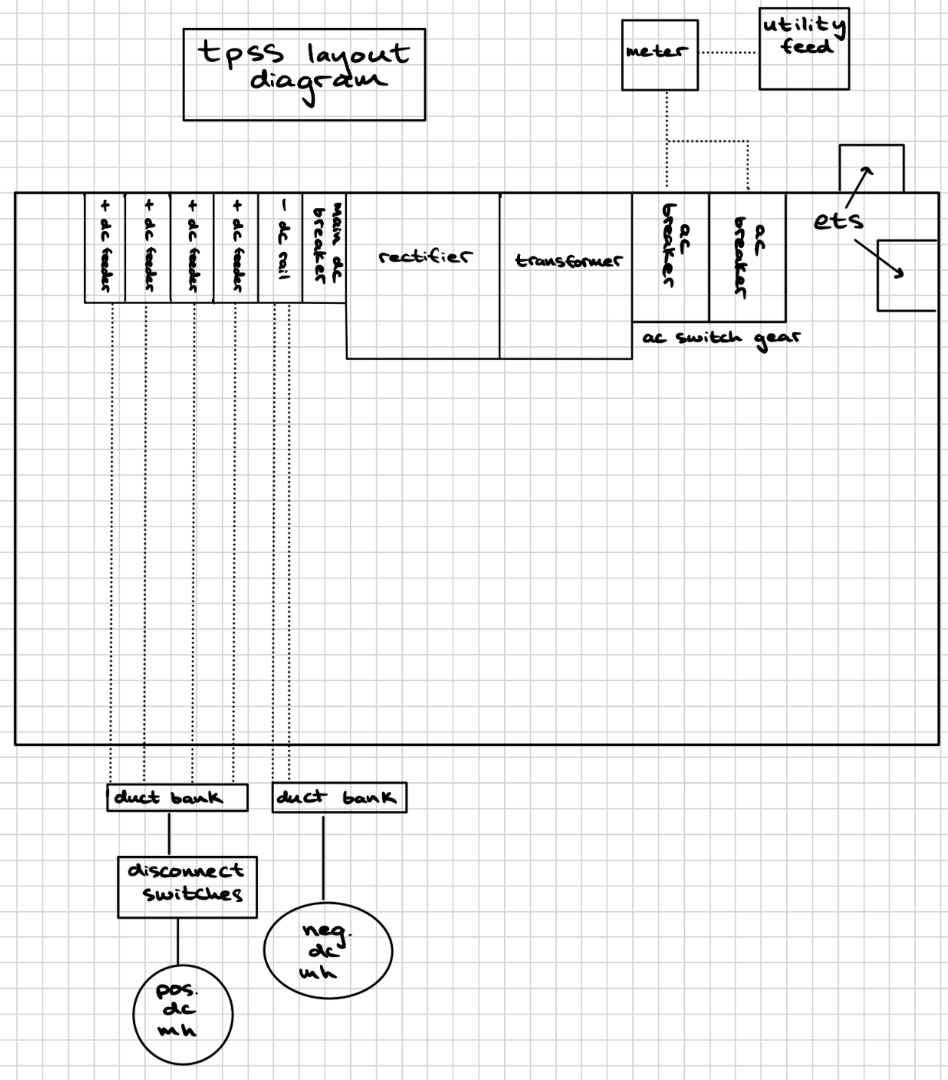
Instrument Transformers

Current Transformer - Measure current to provide input to relays

Voltage Transformers - step down voltage to safe levels to enable voltage monitoring

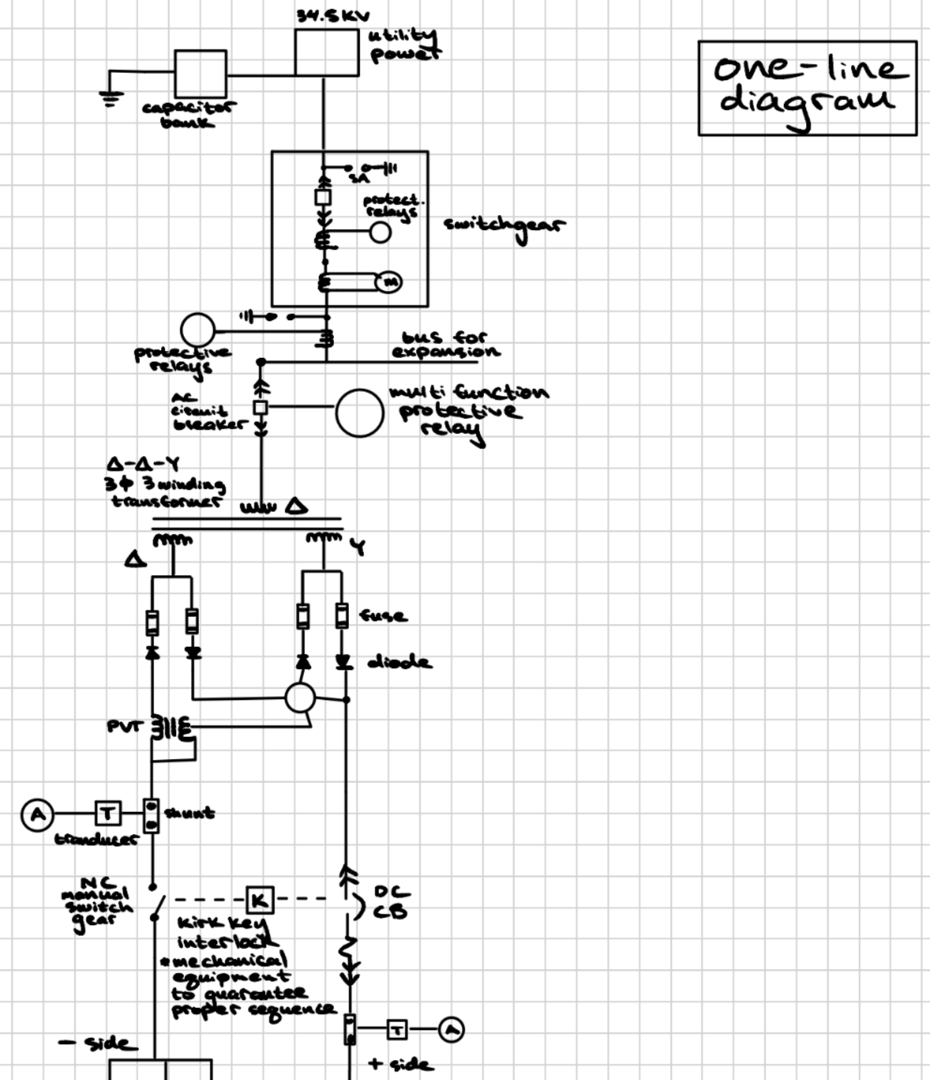
Layout

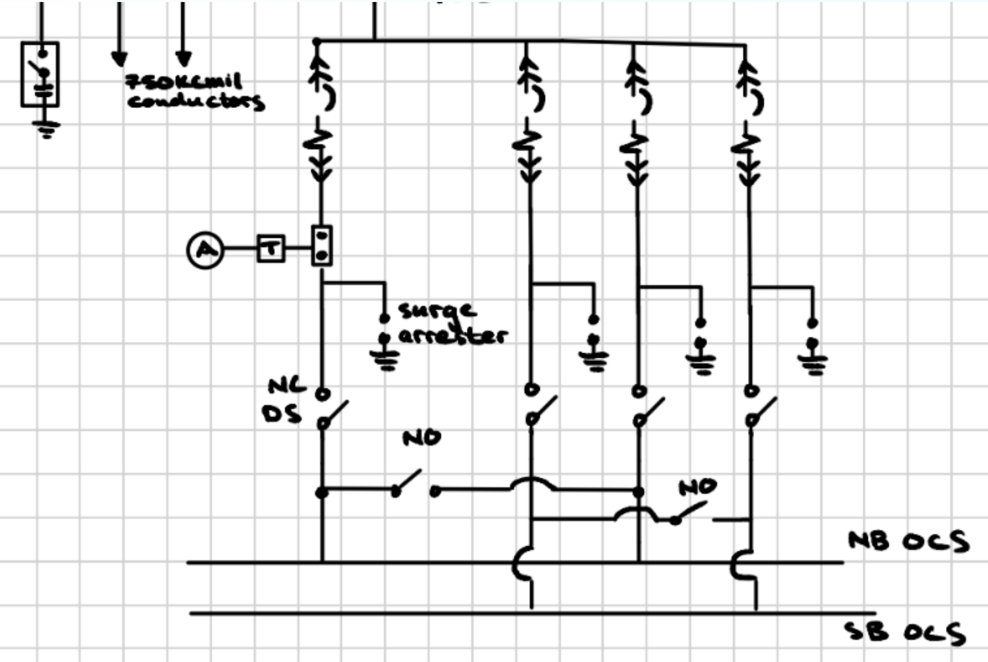
TPSS - protection, rectifier transformer, breakers, etc.

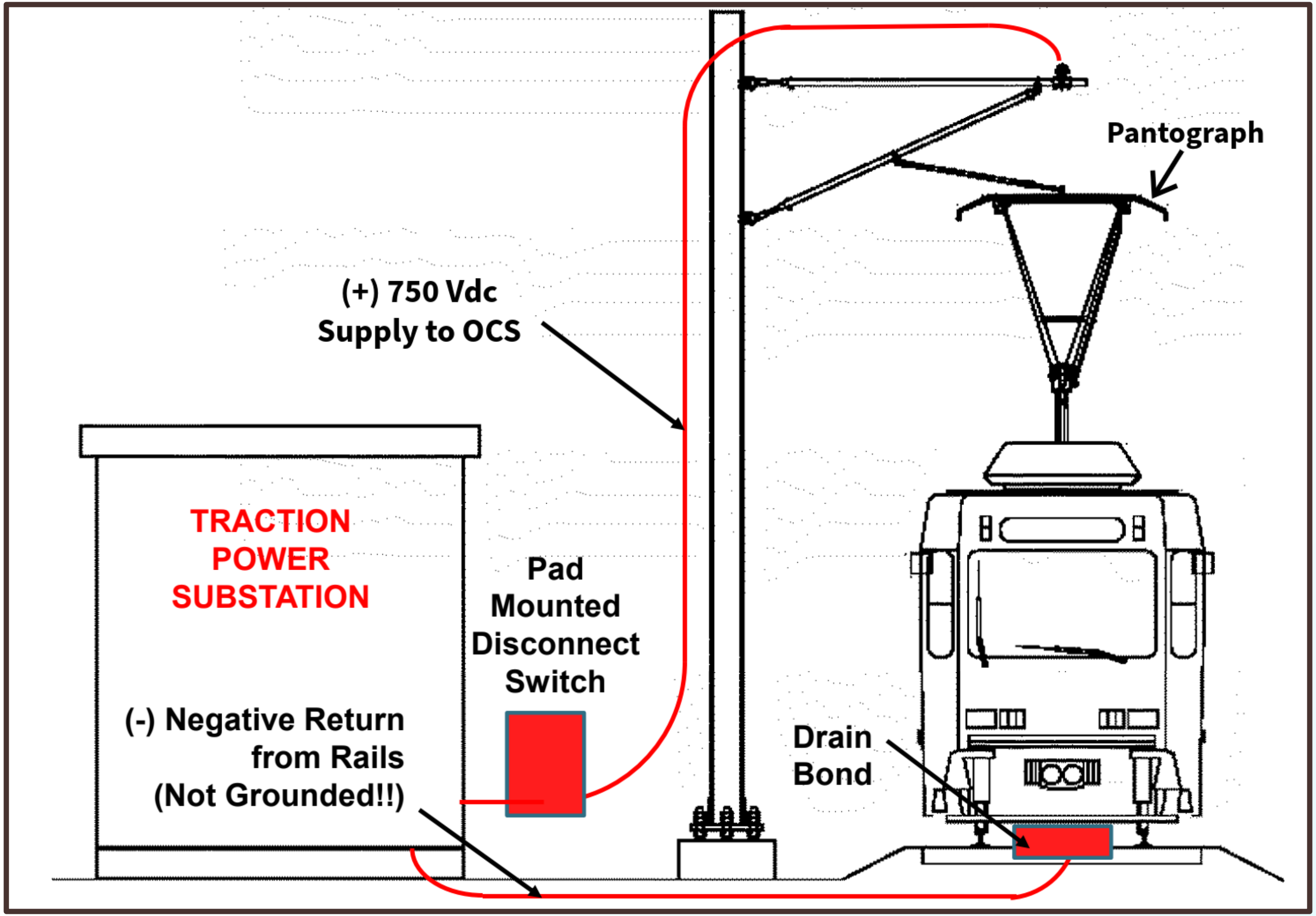


One-Line

Grid to OCS







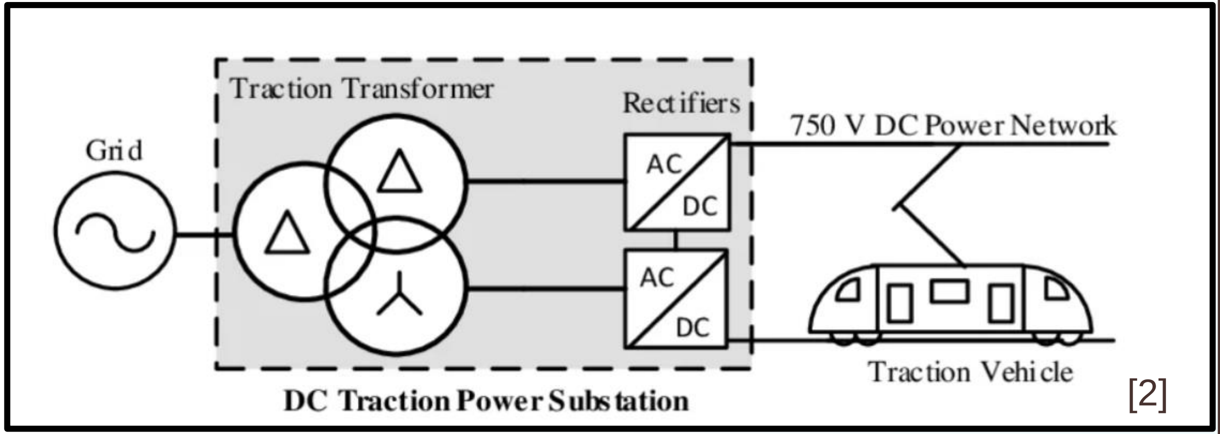
Pantograph
(+) 750 Vdc
Supply to OCS
TRACTION
POWER
SUBSTATION
Pad
Mounted
Disconnect
Switch
(-) Negative Return
from Rails
(Not Grounded!!)
Drain
Bond

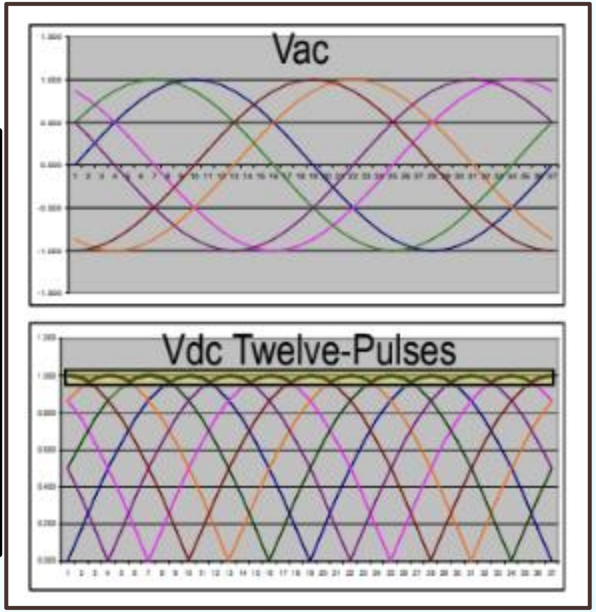
[1]

Solution Voltage/Power Rated Chart

Device	Voltage	Rated Power
Distribution Lines	34.5 KV AC line-line	-
Transformer	34.5 KV AC → 576 V AC	3 MVA
Rectifier	576 V AC → 750 V DC	3 MVA
LRV	750 V DC	-

Rectifier Transformer





[1]

Utility Feed

ROCKY MOUNTAIN POWER®

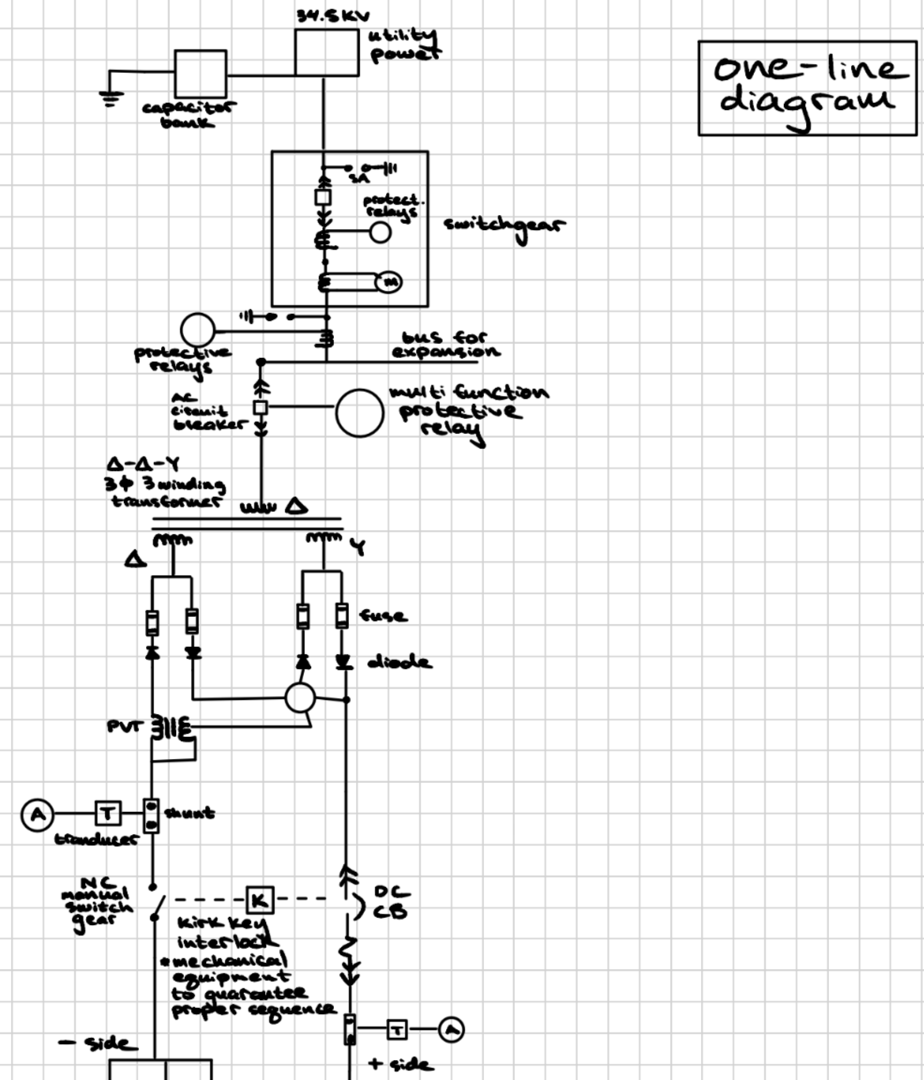
- Coordinate with utility using our 1-line that uses our LRV electrical loads
- In other words, does the utility need money from us to upgrade or build a new transmission line to power our load OR is there an existing utility line there that can supply us 34.5 kV?
- We assumed the existing utility is adequate

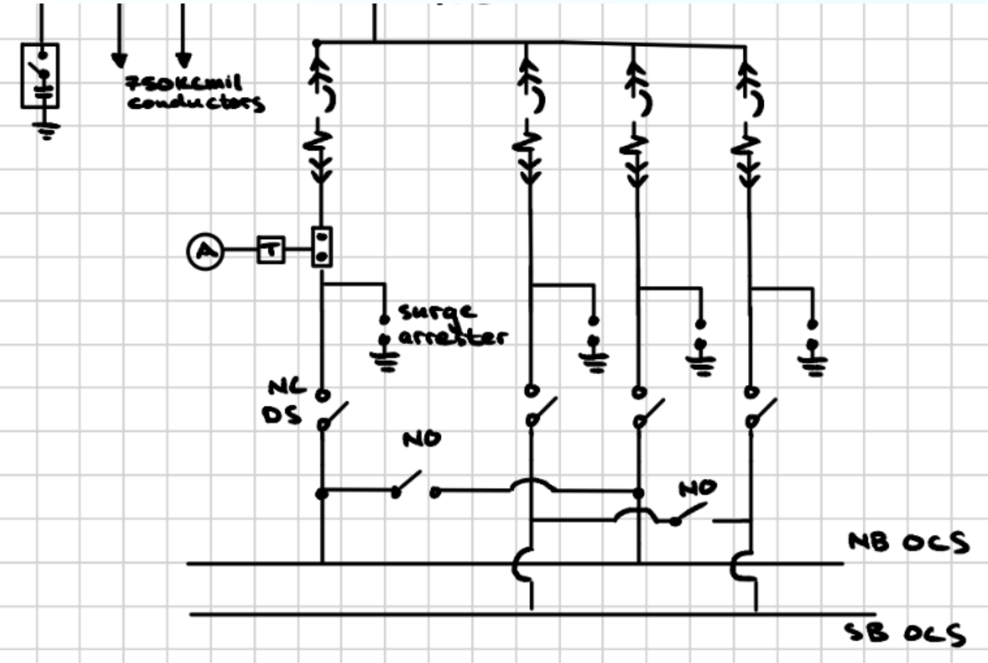
Raceway/Cable Schedule

CONDUIT NO.	FROM	TO	PURPOSE	CONDUIT			CONDUCTOR		
				QUANTITY (EA)	SIZE (IN)	TYPE	QTY/EA	SIZE	TYPE
TP01-HV-01	METERING SWITCHGEAR	TPSS MAIN AC BREAKER	34.5 kV UTILITY FEEDER	1	5	PVC	3/C	TBD	35 kV
TP01-LV-01	TPSS AC DIST PANEL	METERING SWITCHGEAR	SWITCHGEAR LV POWER	1	1	PVC	-	-	600 V
TP01-LV-1	DC BREAKER 1	DC DISCONNECT SWITCH 1	POS FDR TRACK 1 NORTH	1	4	PVC	-	-	-
TP01-POS-1	DC DISCONNECT SWITCH 1	DC POS MANHOLE TEMH 1	POS FDR TRACK 1 NORTH	1	4	PVC	3	750 KCMIL	2 KV, 90°C
TP01-LV-2	DC BREAKER 2	DC DISCONNECT SWITCH 2	POS FDR TRACK 2 NORTH	1	4	PVC	-	-	-
TP01-POS-2	DC DISCONNECT SWITCH 2	DC POS MANHOLE TEMH 2	POS FDR TRACK 2 NORTH	1	4	PVC	3	750 KCMIL	2 KV, 90°C
TP01-LV-3	DC BREAKER 3	DC DISCONNECT SWITCH 3	POS FDR TRACK 1 SOUTH	1	4	PVC	3	750 KCMIL	2 KV, 90°C
TP01-POS-3	DC DISCONNECT SWITCH 3	DC POS MANHOLE TEMH 3	POS FDR TRACK 1 SOUTH	1	4	PVC	3	750 KCMIL	2 KV, 90°C
TP01-LV-4	DC BREAKER 4	DC DISCONNECT SWITCH 4	POS FDR TRACK 2 SOUTH	1	4	PVC	3	750 KCMIL	2 KV, 90°C
TP01-POS-4	DC DISCONNECT SWITCH 4	DC POS MANHOLE TEMH 4	POS FDR TRACK 2 SOUTH	1	4	PVC	3	750 KCMIL	2 KV, 90°C
TP01-SWG 1	SCITC	PAD MTD DISCONNECT SWITCHES	SWITCH CONTROL AND INDICATIONS	1	2	PVC			
TP01-NR 1	NEG BUS	DC NEG MANHOLE TEMH 1	TRACK 1 NEGATIVE RETURNS	1	4	PVC	2	750 KCMIL	2 KV, 90°C
TP01-NR 1	NEG BUS	DC NEG MANHOLE TEMH 1	TRACK 1 NEGATIVE RETURNS	1	4	PVC	2	750 KCMIL	2 KV, 90°C
TP01-NR 2	NEG BUS	DC NEG MANHOLE TEMH 2	TRACK 2 NEGATIVE RETURNS	1	4	PVC	2	750 KCMIL	2 KV, 90°C
TP01-NR 2	NEG BUS	DC NEG MANHOLE TEMH 2	TRACK 2 NEGATIVE RETURNS	1	4	PVC	2	750 KCMIL	2 KV, 90°C

One-Line

Grid to OCS





ETS System

Emergency Trip System (ETS) and Blue Light Station (BLS)

03 Conclusion

Future Design and Construction

Ground Grid

Designing the Ground Grid

Full Design

Auxillary Power
Communications
PLCs
Battery Storage

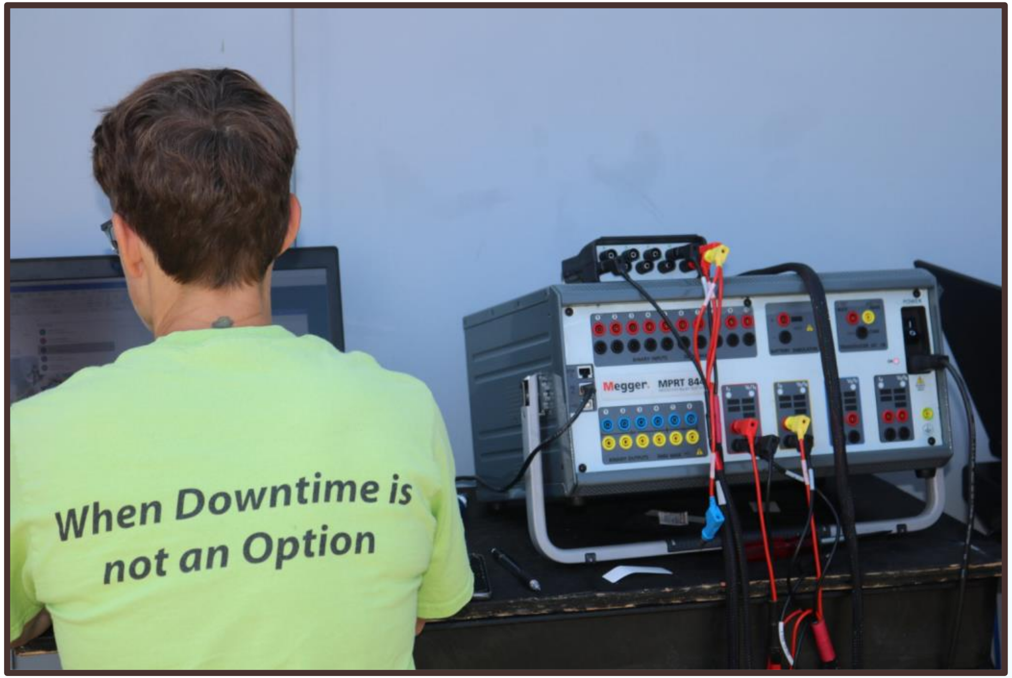
TPSS Fabrication

Sending the full specifications to manufacturer

Excavation of Ground Grid

Foundation is correct & Structural Engineer approved

Testing



- Check and tune protective relays and breakers
- Confirm all control circuits are functional
- Test alarms
- Ensure connection with SCADA (Supervisory Control and Data Acquisition) to ROC (Rail Operation Center)

Delta Center TPSS Video

[3]

References

[1] Ohm Gurus Presentation

[2] Image of rectifier transformer

[3] Facebook RideUTA how to power a trax train

[4] Interview with Experts in Transit